

Exploring individual and community-level variation in quantifier scales

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1. Introduction : Quantifier Scales

1. Set Theoretic Definitions: (Barwise and Cooper 1981)

$\llbracket \text{most} \rrbracket = \lambda P_{\langle e, t \rangle}. \lambda Q_{\langle e, t \rangle}. |P \cap Q| > |P - Q|$ (more than half)

$\llbracket \text{many} \rrbracket = \lambda P_{\langle e, t \rangle}. \lambda Q_{\langle e, t \rangle}. |P \cap Q| > n$ (more than n)

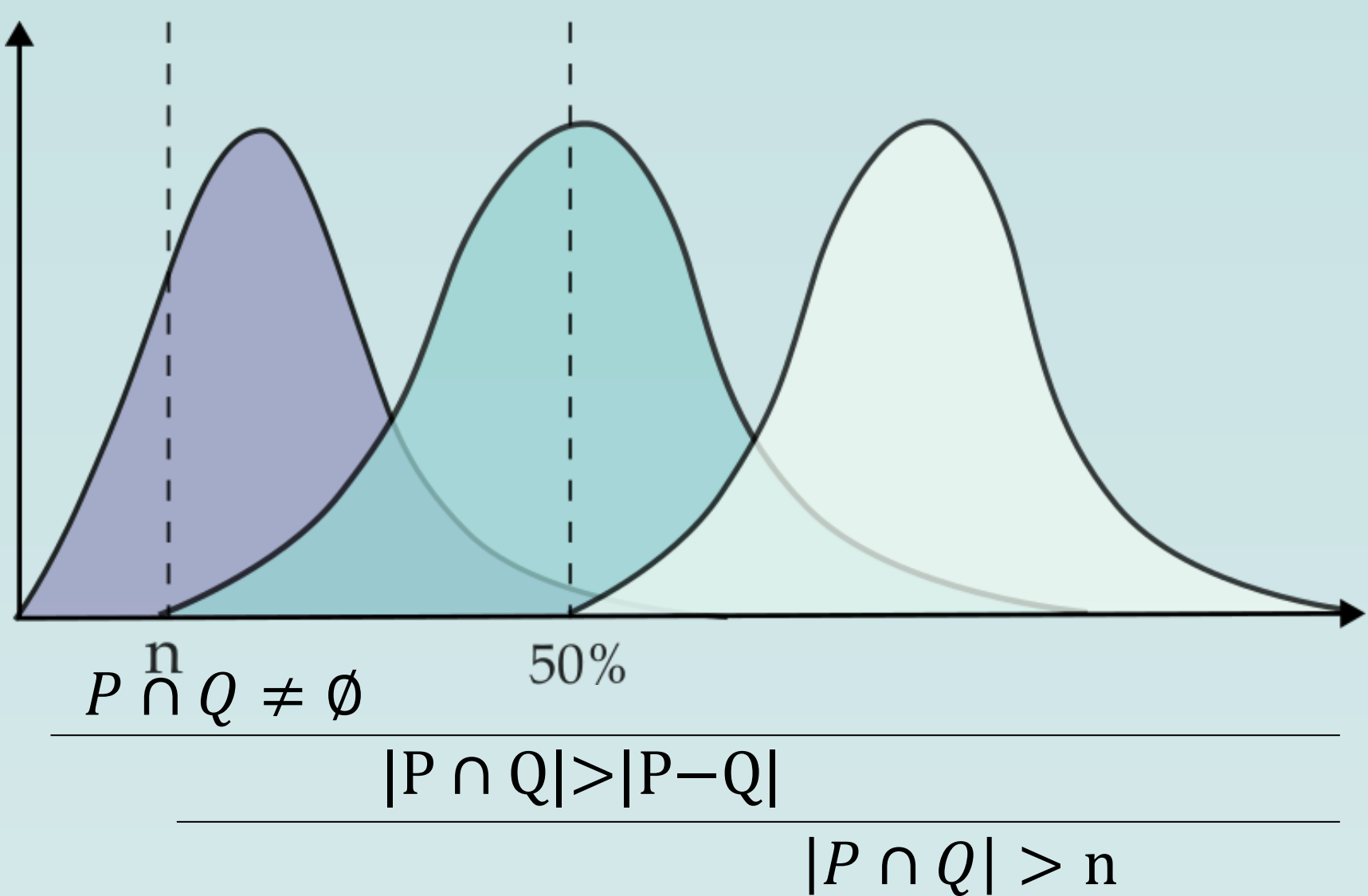
$\llbracket \text{some} \rrbracket = \lambda P_{\langle e, t \rangle}. \lambda Q_{\langle e, t \rangle}. P \cap Q \neq \emptyset$ (at least one)

2. **Entailments:** most $\Rightarrow$ many $\Rightarrow$ some; most $\nRightarrow$ many $\nRightarrow$ some;

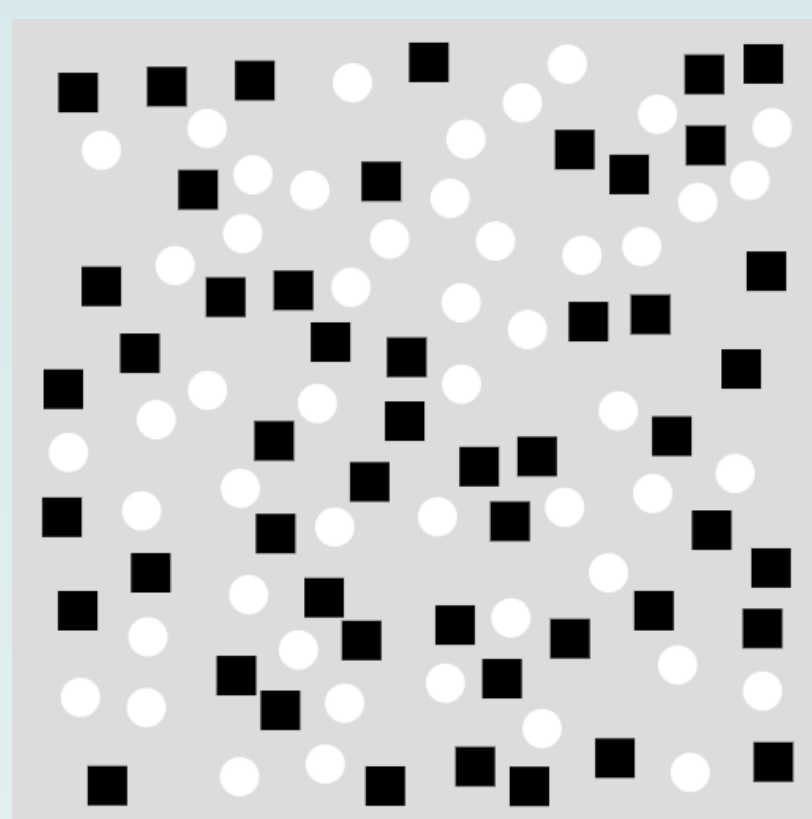
3. **Lexical Scales:** <some, many, most>

4. Scalar Implicature (SI): *not many* (Horn 1972)

Quantifier usage is more restricted than their set theoretic definition



3. Methodology



Choose the option that best describes the image:

[Most/Many/Some/A few] objects are squares.

Approximate proportion of black squares: **[11%-20%, 21%-30%...]**

Spanish: *unos pocos, algunos, muchos, la mayoría*

Catalan: *uns quants*, alguns, molts, la majoria*

Italian: *pochi, alcuni, molti, la maggior parte*

*In Catalan *uns pocs* shows variability in the acceptability judgements
40 native speakers from each language recruited via Prolific. The study was conducted online in PCIBex Farm (Zehr & Schwarz 2018)

5. Discussion

Variation in Quantifiers at the population level:

- In **Catalan** *uns quants* and *some* used in similar contexts
→ a. $\llbracket \text{uns quants} \rrbracket = \llbracket \text{alguns} \rrbracket$; <uns quants/alguns, molts>
- In **Spanish** and **Italian** *unos pocos* and *pochi* are used to represent a smaller magnitude than *algunos* and *alcuni*
→ b. $\llbracket \text{unos pocos} \rrbracket \neq \llbracket \text{algunos} \rrbracket$; <unos pocos, algunos, muchos>
→ b. $\llbracket \text{pochi} \rrbracket \neq \llbracket \text{alcuni} \rrbracket$; <pochi, alcuni, molti>

Variation at the individual level:

There is considerable amount of intra-speaker variation (especially in the case of Catalan) and this could indicate that we are dealing with a case of **covert representational variation** (MacKenzie 2019).

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2. Research Questions

1. How does the quantifier *a few* enter into this picture?

Different analyses for *a few* in the literature (Solt 2006):

- a. $\llbracket \text{a few} \rrbracket = \lambda P_{\langle e, t \rangle}. \lambda Q_{\langle e, t \rangle}. P \cap Q \neq \emptyset$ Upper boundary Scalar Implicature
- b. $\llbracket \text{a few} \rrbracket = \lambda P_{\langle e, t \rangle}. \lambda Q_{\langle e, t \rangle}. |P \cap Q| < n$ Upper boundary lexically encoded

Entailments: a few $\Rightarrow$ some;

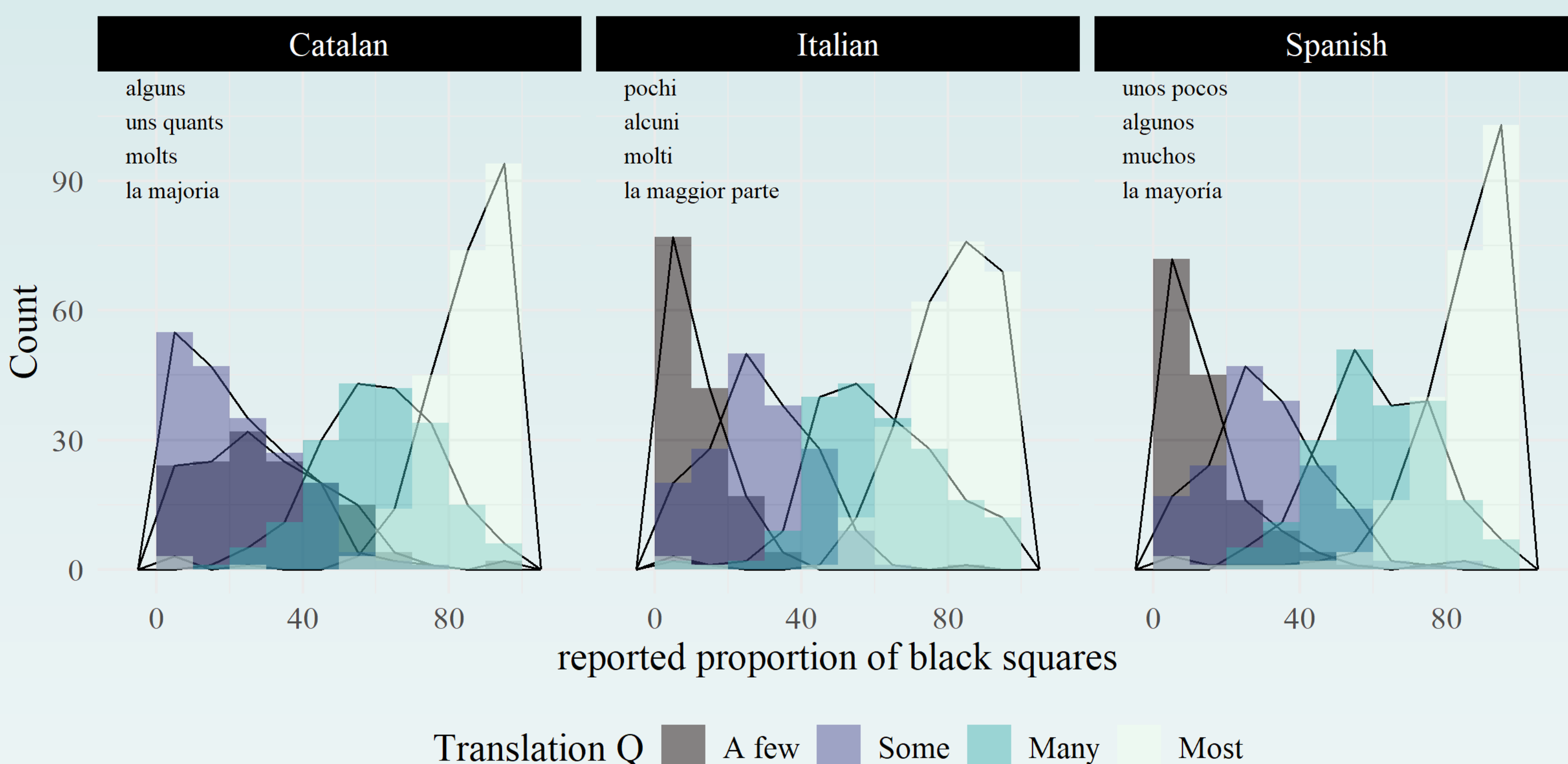
- a. some $\Rightarrow$ a few Should not form a scale
- b. some $\nRightarrow$ a few Should form a scale: <a few, some>

2. How does the space of quantifier alternatives vary across different Romance languages and speakers?

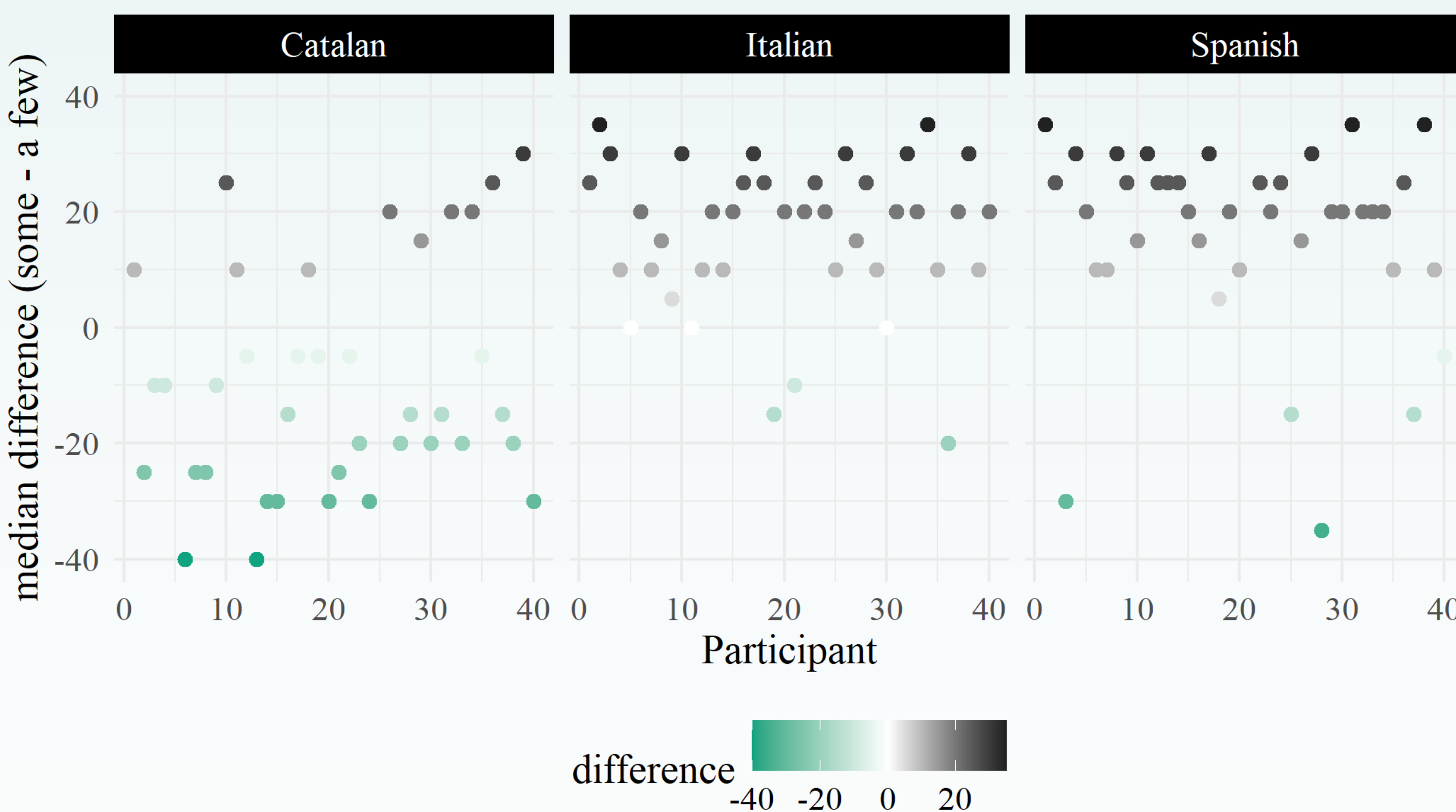
- Individual level variation in SI (Khorsheed and Gotzner 2023)
- Differences across quantifiers in terms of SI (McNabb 2015)
- Variation in children (Pagliarini et. al 2018)
- Quantifiers' alternatives vary diachronically (Bech 2024)

4. Results

Variation at the population level:



Variation at the individual level:



References

- Barwise, J. and Cooper, R. (1981). Generalized quantifiers and natural language. *Philosophy, language, and artificial intelligence: Resources for processing natural language*, 241–301 • Bech, K. (2024). The Old English quantifiers *fela* 'many' and *manig* 'many', and Ælfric as a linguistic innovator. *Noun phrases in early Germanic language* • Horn, L. R. (1972). *On the semantic properties of logical operators in English*. PhD thesis, University of California • Khorsheed, A., & Gotzner, N. (2023). A closer look at the sources of variability in scalar implicature derivation: a review. *Frontiers in Communication*, 8 • MacKenzie, L. (2019). Perturbing the community grammar: Individual differences and community-level constraints on sociolinguistic variation. *Glossa*, 4(1) • McNabb, Y. (2015) Differences and similarities between scalar inferences and scalar modifiers: The case of quantifiers. *SALT*, 267-287 • Pagliarini, E., Bill, C., Romoli, J., Tieu, L., & Crain, S. (2018). On children's variable success with scalar inferences: Insights from disjunction in the scope of a universal quantifier. *Cognition*, 178, 178-192 • Solt, S. D. (2006). Why *a few*? and why not **a many*? *Proceedings of SuB* (10), 333–346 • Zehr, J. and Schwarz, F. (2018). Penncontroller for Internet Based Experiments (IBEX)